

QA CHATBOT API

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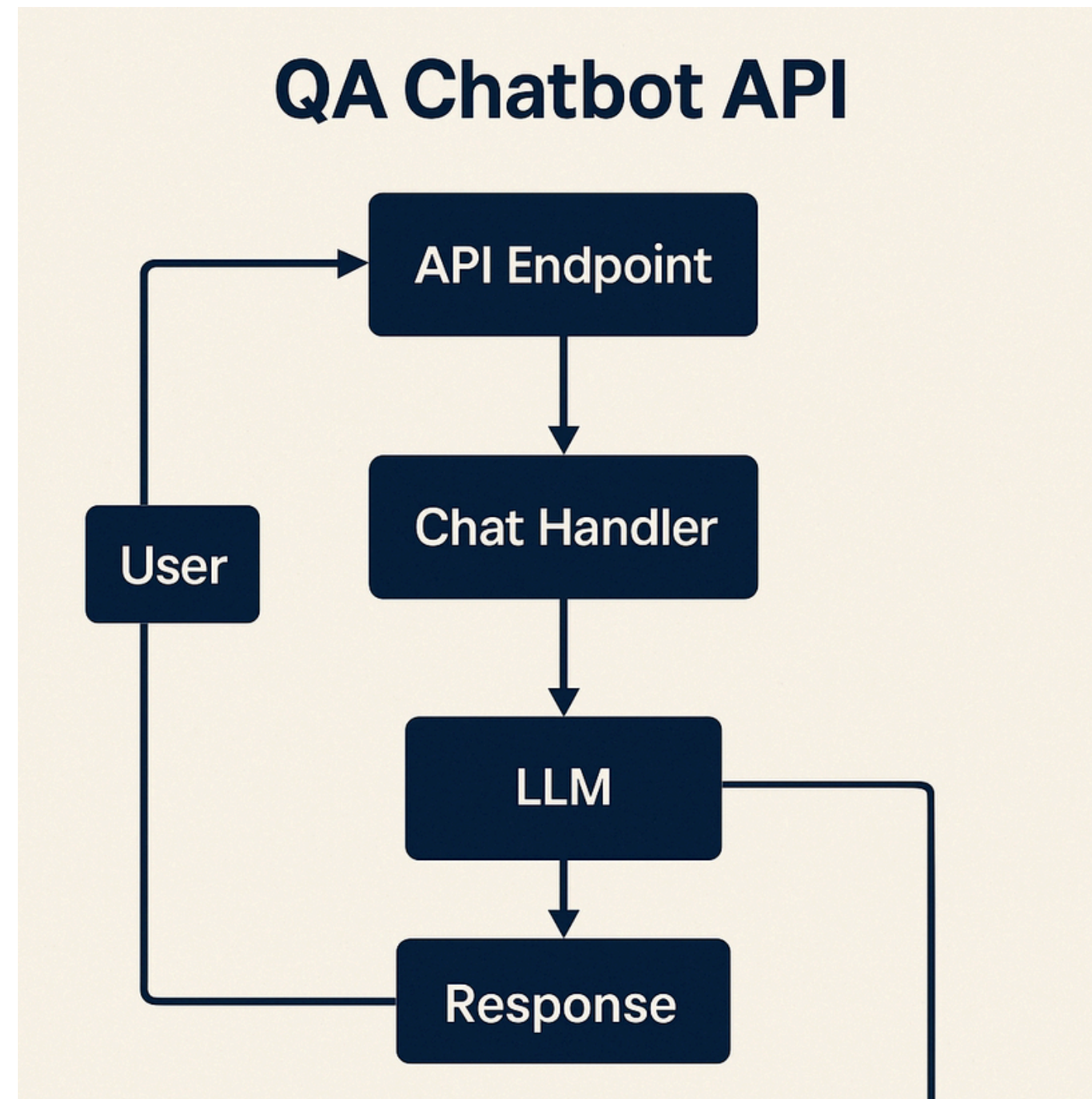
PROJECT OVERVIEW

- What it does:
 - REST API that answers user questions using Gemini AI
 - Stores conversation history in SQLite
 - Validates inputs and handles errors
- Key Features:
 - **/ask** endpoint for questions
 - **/history** endpoint to view past conversations
 - Health check endpoint

TECH STACK

- **Backend:** FastAPI (Python)
- **AI Model:** Google Gemini 1.5 Flash
- **Database:** SQLite (via SQLAlchemy ORM)
- **Testing:** Pytest + FastAPI TestClient
- **Deployment:** (Mention if deployed on Render/other platforms)

ARCHITECTURE DIAGRAM



IMPLEMENTATION CHOICES

Why FastAPI?

- Async support for LLM calls
- Automatic docs (Swagger UI)
- Easy input validation with Pydantic

Why Gemini 1.5 Flash?

- Cost-effective for QA use cases
- Good balance of speed/accuracy

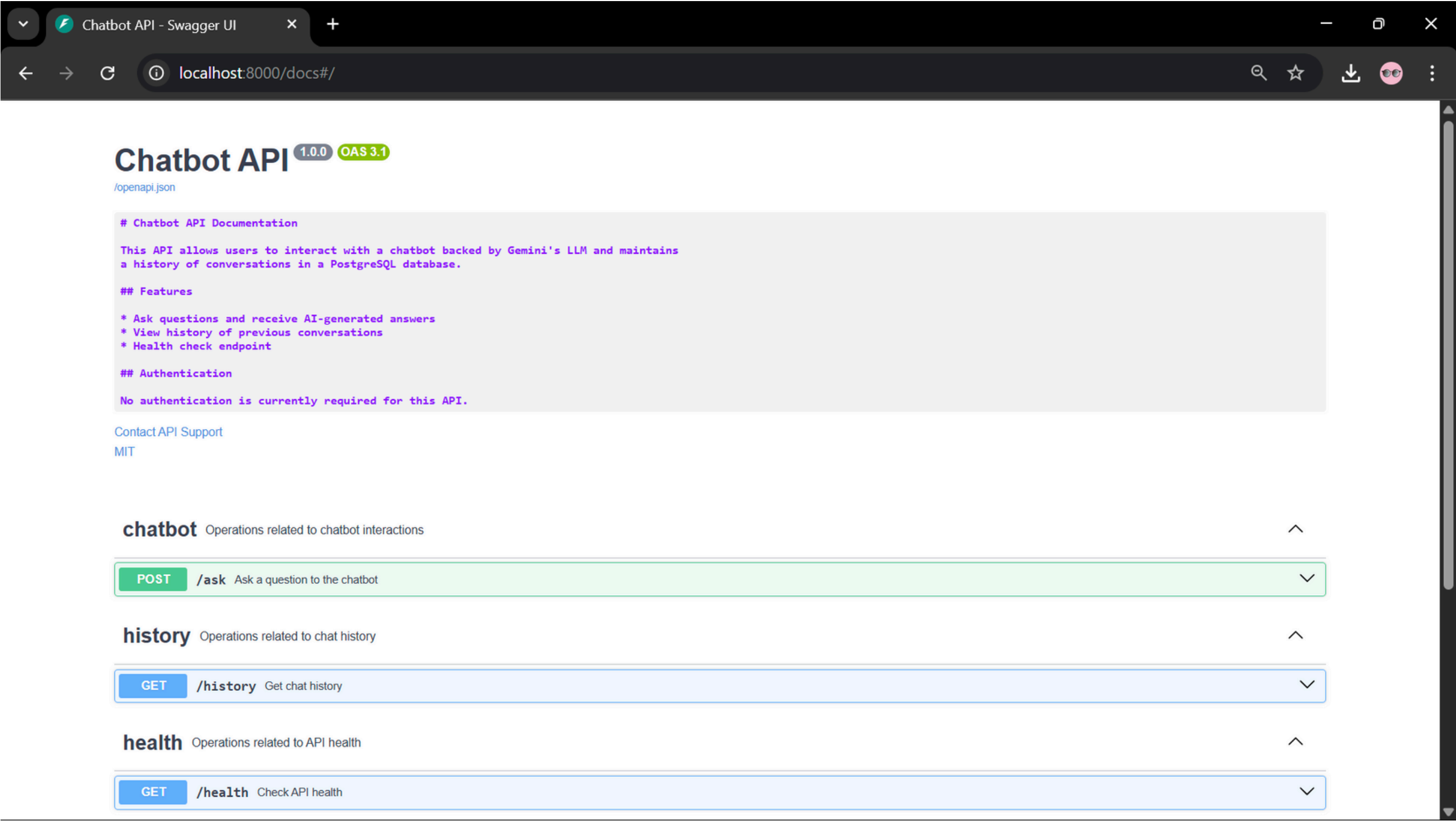
Why SQLite?

- Lightweight for local development
- Easy to migrate to PostgreSQL if needed

CODE STRUCTURE

```
qa-chatbot-api/  
├── app/  
│   ├── __pycache__/  
│   ├── chat.py          # Core chat functionality  
│   ├── llm.py           # Language model integration  
│   └── main.py          # Main application entry point  
├── include/             # Additional resources  
├── lib/                 # Library dependencies  
├── scripts/            # Utility scripts  
├── test/  
│   ├── __pycache__/  
│   ├── .pytest_cache/  
│   └── test_chat.py     # Test cases for chat functionality  
├── venv/               # Virtual environment (not tracked in git)  
├── .env                # Environment variables (not tracked in git)  
├── list.py             # Utility functions  
├── pyenv.cfg           # Python environment configuration  
├── requirements.txt    # Project dependencies  
└── README.md           # This file
```

DEMO



Swagger UI

ASK ENDPOINT

POST

/ask Ask a question to the chatbot

⌵

Send a question to the chatbot and receive an AI-generated answer. The question-answer pair will be stored in the chat history.

Parameters

Cancel

Reset

No parameters

Request body required

application/json

```
{
  "question": "Who is Virat Kohli?"
}
```

The ASK endpoint allows users to send a natural language question to the chatbot via a POST request. This is visually presented through Swagger UI for interactive testing and documentation.

Response body

```
{
  "question": "Who is Virat Kohli?",
  "answer": "Virat Kohli is a highly accomplished Indian cricketer, widely regarded as one of the greatest batsmen of all time. He's primarily known for his aggressive batting style and exceptional record in all formats of the game (Test, ODI, and T20I). He's captained the Indian national cricket team across all formats and has led numerous successful campaigns. Beyond his on-field achievements, he's also a highly popular and influential figure in India and globally, with significant endorsements and a large fanbase.\n",
  "success": true,
  "error": null
}
```

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HISTORY ENDPOINT

Response body

```
[
  {
    "id": 8,
    "question": "Who is Virat Kohli?",
    "answer": "Virat Kohli is a highly accomplished Indian cricketer, widely regarded as one of the greatest batsmen of all time. He's primarily known for his aggressive batting style and exceptional record in all formats of the game (Test, ODI, and T20I). He's captained the Indian national cricket team across all formats and has led numerous successful campaigns. Beyond his on-field achievements, he's also a highly popular and influential figure in India and globally, with significant endorsements and a large fanbase.\n",
    "timestamp": "2025-05-10T11:46:55.981960"
  },
  {
    "id": 7,
    "question": "what is PSL",
    "answer": "PSL can stand for several things, depending on the context. The most common meanings are:\n\n* **Pakistan Super League:** A professional Twenty 20 cricket league in Pakistan. This is probably the most common meaning in sports-related contexts.\n\n* **Pumpkin Spice Latte:** A popular seasonal coffee beverage, typically available in the fall at coffee shops like Starbucks. This is likely the most common meaning in a food and beverage context.\n\nWithout more context, it's impossible to say for sure which meaning is intended.\n",
    "timestamp": "2025-05-10T02:51:50.122130"
  },
  {
    "id": 6,
    "question": "what is IPL",
    "answer": "IPL can refer to a few things, most commonly:\n\n* **Indian Premier League (IPL):** This is a professional Twenty20 cricket league in India. It's one of the most popular and lucrative T20 leagues in the world, featuring top international and Indian players.\n\n* **Intense Pulsed Light (IPL):** This is a non-invasive cosmetic procedure that uses broad spectrum light to treat various skin conditions such as hair removal, acne, rosacea, and age spots.\n\nIt's important to consider the context to understand which meaning of IPL is being used.\n",
    "timestamp": "2025-05-10T02:51:30.113920"
  }
]
```

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Displays a list of past Q&A pairs with id, question, answer, and timestamp.
Useful for tracking conversation history and understanding user queries over time.

TESTING

```
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
```

```
===== 10 passed, 12 warnings in 7.85s =====
```

- Pytest was used for validating core functionality.
- All tests passed successfully with key checks for:
 - LLM response handling (success & failure)
 - Mocked database chat history
 - Input validation (e.g., empty or long questions)

FUTURE IMPROVEMENTS

- 1. Enhanced Security
 - JWT/OAuth2 Integration
 - Secure endpoints with token-based auth
 - /login endpoint + role-based access control
- 2. Cloud Deployment
 - Options: Render (Free Tier) or AWS (EC2 + RDS)
 - Dockerize + CI/CD pipeline
 - Auto-scaling for high traffic
- 3. Multi-Model Support
 - Fallback Strategy:
 - Primary: Gemini 1.5 Flash → Fallback: OpenAI GPT-3.5
 - Cost-effective & fault-tolerant
- 4. History Management
 - New Endpoints:
 - DELETE /history/{id} (Single entry)
 - DELETE /history/all (Admin-only)



THANK YOU